Raghav Prabhakar

Applied Machine Learning Engineer — MSc AI (VU Amsterdam)
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Summary

Machine Learning Engineer with 2 years of experience building and deploying large-scale AI systems across speech, generative AI, and multimodal applications. Experienced in dataset curation at scale (150k+ hours), fine-tuning speech and language models, and developing production-grade ML pipelines. Proven track record with open-source systems (Flowpilot ADAS - 2000+ GitHub stars) and real-world deployments.

Education

Vrije Universiteit — Amsterdam, Netherlands
Masters of Science, Artifical Intelligence — *September. 2025 - Present*

Thapar Institute of Engineering and Technology — Patiala, India
Bachelors of Engineering, Computer Engineering, CGPA: 8.6/10 — *2019 - 2023*

Experience

Machine Learning Engineer — December. 2023 – August. 2025
Dubverse.ai — *Gurugram, India*

- Built and deployed speech and audio ML systems including language classification, transcription, and music separation to enhance product capabilities
- Designed and scaled multilingual Text-to-Speech pipelines supporting 60+ languages and 500+ voices, achieving 23.8× faster inference compared to baseline systems   Improved inference performance and system efficiency for
- Led a team of 2–3 engineers, improving system scalability and optimizing infrastructure costs for large-scale model training and production-scale speech generation and dubbing system.
- Built internal ML pipelines and analytics systems to track usage (API calls, character volume, speaker distribution, dataset annotation, model evaluation) and improve model performance

Research Assistant — September. 2023 – February. 2024
National University Of Singapore (NUS), Singapore — *Remote*

- Developed frameworks and Question-Answering datasets to analyse and improve physical common-sense reasoning in embodied agents thus improving reasoning capabilities of vision-language agents

Research Assistant — January. 2023 – November. 2023
Robotics Research Centre (RRC) and Machine Learning Lab (MLL), IIIT Hyderabad — *Hyderabad, India*

- Developed pipelines for instance-level semantic mapping for vision-language navigation.
- Built gesture-controlled robot with real-time tracking and obstacle avoidance.
- Explored LLM-based control systems for embodied agents and code generation.
- Built pose estimation and tracking systems for real-world fitness applications and deployed models in production gym environments for user feedback and performance tracking

Co-Founder — April. 2020 – June. 2023
Flowdrive.ai — *Patiala, India*

- Architected and developed an open-source, cross-platform Level 2 Advanced Driver Assistance System (ADAS)
- Developed pseudo LIDAR system using cameras and deep learning, utilizing DenseDepth for 3D reconstruction from monocular images.
- Implemented deep learning-based behavioural cloning models for autonomous vehicles, creating models that learn and imitate human driving patterns.

Publication

1. A. Agrawal, **Raghav Prabhakar**, A. Goyal, and D. Liu. Physical Reasoning and Object Planning for Household Embodied Agents. *Transactions on Machine Learning Research*, 2024
2. K. M. Krishna, L. Nanwani, A. Agarwal, K. Jain, **Raghav Prabhakar**, A. Monis, A. Mathur, K. Murthy, A. Hafez, and V. Gandhi. Instance-Level Semantic Maps for Vision Language Navigation. In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, pages 507–512, 2023
3. J. Singh, A. R. Chowdhury, **Raghav Prabhakar**, and V. C. W. MahaTTS: A Unified Framework for Multilingual Text-to-Speech Synthesis. Technical report, 2025

Projects

MahaTTS | *Python, Pytorch, React, PostgreSQL, Docker, Django*

- Open-source multilingual TTS (270+ stars) based on TortoiseTTS with wav2vec2 tokens, trained on 20K+ hours of data. Built an internal production version trained on 150K+ hours using 32× A100 GPUs. Developed full-stack scalable deployment system for audiobook generation at scale.

Flowpilot | *Python, C, Java, Docker*

- Worked on Flowpilot, an open-source driver assistance system (2000+ stars) built on openpilot enabling Adaptive Cruise Control (ACC), Automated Lane Centering (ALC), Forward Collision Warning (FCW) etc across Windows, Linux, and Android devices for multiple community-supported car models.

Embodied Evolution via Spatial Mating | *Python*

- Built an evolutionary robotics system where all robots exist and interact in a shared environment, enabling spatial mating instead of isolated evaluations. Studied mate selection strategies and their impact on fitness and population diversity through simulation.

Technical Skills

Languages: Python, Java, C++, SQL
Frameworks: Pytorch, Transformers, ROS Navigation Stack, Flask, Celery, FastAPI
Developer Tools: Git, Docker, GCP, AWS, Azure, Gradle, Swagger, Grafana, Prometheus

Achievements

Top 8% in Kaggle Mayo Clinic - STRIP AI Competition
OpenAI Researcher Access Program, 2023